

CS 25000 Lab 11 – upload by the end of your lab session. Work in a team of two or three.

Instructions for each homework assignment.

- Delete nothing from this Word file.
- Edit this file to add your typewritten answers to each question.
- Ensure that your answer fits on the same page as its question. If you change the pagination of this file or if your complete answer to a question does not fit on the page with that question, then you may receive a lower score.
- When your answer includes a hand-drawn diagram make it clear, large, and legible.
- Export your completed Word file to PDF.
- Upload your PDF file to its corresponding assignment portal in Gradescope.com. You may upload multiple times. Your final upload will be scored. As desired, use the download capability to check your upload.
- Upload by the end of your lab session.
- Max score = 20 points; 2 points per question.

- Why should your answer be on the same page with its question?

Answer: Gradescope has been programmed to expect that your PDF file will have exactly 10 pages, each containing a question and your entire answer. This allows Gradescope to automatically display each answer for scoring.

1. Why is an affordable amount of SRAM memory that merely stores a copy of information that is also present in the DRAM main memory able to improve the execution time of most programs?

Memory from DRAM that needs to be accessed frequently can be copied into SRAM, where it can be accessed faster, giving the illusion that the program has faster memory overall.

2. An SRAM array is square and stores 2^{20} bits. How many word lines does the array contain? Why?

1024 word lines

A 2^{20} bit square is $2^{10} \times 2^{10}$ bits. Thus, the array contains 1024 word lines.

3. An SRAM array is square and stores 2^{20} bits. How many 64-bit words stored in the array could be accessed simultaneously?

1024 bits are read out at once

$1024 \text{ bits} / 64 \text{ bits / word} = 16 \text{ words}$

16 64-bit words can be accessed simultaneously

4. Assume a computer has a physical memory organized into 8-byte words. Using hexadecimal notation, give the word address and offset within the word for each of the following byte addresses. Left pad with the minimum number of zeros necessary to make each bit string in your answers a multiple of 4 bits in size.

Answer:

Byte address	Word address	Offset
0x001c	0x 0003	0x 0004
0x02ad	0x 0055	0x 0005

5. If a computer has a 48-bit memory address bus, and each address corresponds to one byte, and if DDR5-5600 2x16GiB costs US\$ 6.25/Gibibyte (late Oct. 2025), what is the size of this 48-bit address space in units of gibibytes and in power of 2 form, and what is the approximate cost today for the amount of DRAM to populate this space?

State the DRAM cost rounded to three significant digits and in units of US\$ *value* $\times 10^6$ (i.e., in units of millions of dollars).

Total addressable space = 2^{48} bytes

Address space = 2^{48} bytes / 2^{30} bytes / GiB = **2^{18} GiB**

Total cost = 2^{18} GiB * \$6.25 / GiB = \$1,638,400.00 = **1.638 million dollars**

6. A computer with 32-bit byte-addressed memory has a direct-mapped cache with 512 blocks and 18-bit tags. How many bytes are there in each cache block?

Number of blocks = $2^{\text{(index bits)}}$

$$512 = 2^9$$

9 index bits

Total address bits = 32

Total address bits = tag bits + index bits + offset bits

$$32 = 18 + 9 + \text{offset bits}$$

Offset bits = 5

Bytes per block = $2^{\text{(offset bits)}}$

$$2^5 = \mathbf{32 \text{ bytes per block}}$$

7. If the cache hit rate for a given program is 80%, the hit time for the cache is 10 nanoseconds, and the cache miss time is 1000 nanoseconds, what is the average memory access time for this program, and what is the overall speedup as a percentage from this cache for this program according to Amdahl's Law?

$$\begin{aligned} \text{AMAT} &= \text{Hit time} + \text{Miss rate} * \text{Miss penalty} \\ &= 10 + .20 * 1000 = \mathbf{210 \text{ nanoseconds}} \end{aligned}$$

Without cache, every access takes 1000 ns

$$(1000-210) / 1000 = \mathbf{79\% \text{ speedup}}$$

8. The following average memory access time (AMAT) equation is a model for performance of a two-level memory hierarchy comprising a cache and a memory.

$$\text{AMAT} = \text{Hit Time} + \text{Miss Rate} \times \text{Miss Penalty}$$

Assume a computer with a two-level memory hierarchy and a C program written without paying any attention to that memory hierarchy.

Which term(s) of the AMAT equation might be improved if the speed of hardware improves?

The hit time and the miss penalty may be improved by faster hardware. The miss rate depends on the program and cache organization, so it will not change.

9. How many direct-mapped cache organizations are possible given 32-bit memory addresses, byte addressing by software, word-addressed physical memory, 4-byte words, and a 17-bit tag? How many wires are needed to transport a block from the DRAM to the SRAM for the cache design with the smallest block size, and how many wires are needed for the design with the largest block size?

Total bits = Tag bits + Index bits + Offset bits

$32 = 17 + \text{Index bits} + \text{Offset bits}$

Index bits + Offset bits = 15

Block size = 2^o (offset bits)

Number of cache blocks = 2^i (index bits)

We have a word size of 4, meaning o must be at least 2.

$2^o \geq 4; o \geq 2$

Thus o can range from 2 to 15, meaning i ranges from 13 to 0 accordingly.

$o_{\min} = 2; i = 13$

$o_{\max} = 15; i = 0$

Counting $o = 2, o = 3, \dots, o = 15$ we get 14 possible values for o meaning **14 possible direct-mapped cache organizations**.

We can calculate 2^o to find the block size in bytes. The number of wires is equal to the number of bits, thus we multiply by 8 bits per byte to get the number of wires needed for the smallest and largest organizations.

Smallest design: $2^2 \text{ bytes} * 8 \text{ bits/byte} = \mathbf{32 \text{ bits} = 32 \text{ wires}}$

Largest design: $2^{15} \text{ bytes} * 8 \text{ bits/byte} = \mathbf{262,144 \text{ bits} = 262,144 \text{ wires}}$

10. Cache misses can be categorized into three groups: compulsory miss, capacity miss, and conflict miss. Assume a cache is initialized to all blocks are invalid. Now execution of a program is started.

- a. Which of these three categories of miss is/are possible if program execution produces an execution address trace where

$$\text{Memory_address_i} = \text{Starting_memory_address} + i \times 4?$$

Compulsory misses only

- b. Which of these three categories of miss is/are possible if the program of Part (a) now includes if or if/else control flow?

Compulsory misses only

- c. After enough program execution time passes, what will be the condition of the valid bits of this cache?

All valid bits in the cache will be valid state

- d. Which of these three categories of miss is/are possible if the program of Part (b) now includes multiple calls to one procedure?

Capacity, and conflict misses are all possible